

PALINDROMIC WIDTH OF FINITELY GENERATED SOLVABLE GROUPS

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ABSTRACT. We investigate the palindromic width of finitely generated solvable groups. We prove that every finitely generated 3-step solvable group has finite palindromic width. More generally, we show the finiteness of palindromic width for finitely generated abelian-by-nilpotent-by-nilpotent groups. For arbitrary solvable groups of step ≥ 3 , we prove that if G is a finitely generated solvable group that is an extension of an abelian group by a group satisfying the maximal condition for normal subgroups, then the palindromic width of G is finite. We also prove that the palindromic width of $Z \wr \mathbb{Z}$ with respect to the set of standard generators is 3.

1. INTRODUCTION

Let G be a group with a set of generators X . A reduced word in the alphabet $X^{\pm 1}$ is a *palindrome* if it reads the same forwards and backwards. The palindromic length $l_P(g)$ of an element g in G is the minimum number k such that g can be expressed as a product of k palindromes. The *palindromic width* of G with respect to X is defined to be $\text{pw}(G, X) = \sup_{g \in G} l_P(g)$. When there is no confusion about the underlying generating set X , we simply denote the palindromic width with respect to X by $\text{pw}(G)$. Palindromes in groups have already proved useful in studying various aspects of combinatorial group theory and geometry, for example see [9]–[13], [15]–[19], [21].

For g, h in G , the *commutator* of g and h is defined as $[g, h] = g^{-1}h^{-1}gh$. If $\mathcal{C}$ is the set of commutators in some group G then the commutator subgroup G' is generated by $\mathcal{C}$. The *commutator length* $l_C(g)$ of an element $g \in G'$ is the minimal number k such that g can be expressed as a product of k commutators. The *commutator width* of G is defined by $\sup_{g \in G'} l_C(g)$ and is denoted by $\text{cw}(G)$. It is well known [22] that the commutator width of a free non-abelian group is infinite, but the commutator width of a finitely generated nilpotent group is finite (see [3, 4]). An algorithm of the computation of the commutator length in free non-abelian groups can be found in [5].

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In [7], we initiated the investigation of finding the palindromic width of a finitely generated group that is free in some variety of groups. We proved that the palindromic width of a finitely generated free nilpotent group is finite. Further using a result of Akhavan-Malayeri and Rhemtulla [2], we proved that $\text{pw}(F_n(\mathcal{UN}), X) \leq 5n$, where $\mathcal{U}$ is the variety of abelian groups, $\mathcal{N}$ is the variety of nilpotent groups and $X = \{x_1, \dots, x_n\}$ is a basis of the group $F_n(\mathcal{UN})$ that is free in $\mathcal{UN}$. In our work we also observed that there is a surprising analogy between commutator widths and palindromic widths in groups, though we did not have a clear picture about the relationship. We asked [7, Problem 2] a question to obtain a connection between the two widths. In this paper we also investigated this question and obtained further results in this direction, see Section 2.

The main aim of this paper is to investigate the palindromic width of some solvable groups and the wreath product of cyclic groups. We found a striking analogy of the palindromic widths and commutator widths of solvable groups. Rhemtulla [23, Theorem 1] proved that if A is a normal abelian subgroup of a solvable group G such that G/A satisfies the maximal condition for normal subgroups, then $\text{cw}(G) < \infty$. The following is an analogous theorem for palindromes.

Theorem 1.1. *Let A be a normal abelian subgroup of a finitely generated solvable group $G = \langle X \rangle$ such that G/A satisfies the maximal condition for normal subgroups. Then $\text{pw}(G, X) < \infty$.*

We recall that a group G is said to satisfy *the maximal condition for normal subgroups* if every normal subgroup of G is the normal closure of a finite subset of G . Our proof of the above theorem is based on the approach of Rhemtulla [23] where he studied commutator width in finitely generated solvable groups. Essentially, we have been able to transfer the language of commutator width into that of palindromes. Rhemtulla [23] also obtained finiteness of the commutator width of free 3-step solvable groups. In this paper, we obtain analogous results for palindromic width of free 3-step solvable groups. We further investigated the palindromic width of free abelian-by-nilpotent-by-nilpotent groups. We prove the following theorem in Section 4.

Theorem 1.2. *Let G be a finitely generated group in the variety $\mathcal{UNN}$. Then the palindromic width of G is finite.*

As a corollary we have:

Corollary 1.3. *Every finitely generated 3-step solvable group has finite palindromic width.*

We do not know whether the palindromic width of an arbitrary finitely generated solvable group (of step > 3) is finite. Note, that an answer on the similar question on the commutator width also unknown.

In [8], we have proved that the palindromic width of a finitely generated free metabelian group of rank n is at most $4n - 1$. It would be interesting to obtain an exact value for the palindromic width of a finitely generated metabelian group. It is indeed possible to obtain a precise value of the palindromic widths for some metabelian groups. In this paper, we consider the special case of the metabelian group $G = \mathbb{Z} \wr \mathbb{Z}$. If a and b are the generators of the first and the second cyclic group in the above

wreath product, then we prove that $\text{pw}(\mathbb{Z} \wr \mathbb{Z}, \{a, b\}) = 3$, see Section 5. Along the way we also show that the commutator width of $\mathbb{Z} \wr \mathbb{Z}$ is 1.

After finishing this work, we learned that Riley and Sale [24] have also studied palindromic width of solvable and metabelian groups using different techniques. They have proved the special case of Theorem 1.1 when A is the trivial subgroup, that is, their result states that the palindromic width of a finitely generated solvable group that satisfies the maximal condition for normal subgroups has finite palindromic width. They have proved this using a result of Akhavan-Malayeri [1]. Riley and Sale have also proved that the palindromic width of $\mathbb{Z} \wr \mathbb{Z}$ is 3. Their proof is based on their estimate of $\text{pw}(G \wr \mathbb{Z}^r)$, where G is a finitely generated group. On the other hand, our proof relies on the fact (proved in Section 5) that any element in the commutator subgroup of $\mathbb{Z} \wr \mathbb{Z}$ is a commutator. Palindromic width in wreath products has also been investigated by Fink [14] who has also obtained an estimate of $\text{pw}(G \wr \mathbb{Z}^r)$. Fink has also obtained the finiteness of palindromic widths of wreath products of some more classes of groups.

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2. COMMUTATOR WIDTH AND PALINDROMIC WIDTH

In this paper we shall use the following result by Rhemtulla [23, Lemma 2].

Lemma 2.1. *Let $G = \langle A, x_1, x_2, \dots, x_n \rangle$ where A is an abelian normal subgroup of G . Then the subgroup $[A, G]$ is precisely the following set*

$$[A, G] = \{[a_1, x_1][a_2, x_2] \dots [a_n, x_n] \mid a_i \in A\}.$$

We shall also use the following theorem by Rhemtulla and Akhavan-Malayeri [2] to obtain a better bound for the metabelian groups.

Theorem 2.2. *Let $G = \langle x_1, \dots, x_n \rangle$ is a non-abelian free abelian-by-nilpotent group of rank n . Let A be an abelian normal subgroup of G such that G/A is nilpotent. Then every element $g \in G'$ can be expressed as:*

$$g = [u_1, x_1]^{a_1} [u_2, x_2]^{a_2} \dots [u_n, x_n]^{a_n}$$

for $u_1, \dots, u_n \in G$ and $a_1, \dots, a_n \in A$, where $u^b = b^{-1}ub$.

Also, we note the following lemma that is often useful to estimate palindromic width.

Lemma 2.3. [7] *Let $G = \langle X \rangle$ and $H = \langle Y \rangle$ be two groups, $\mathcal{P}(X)$ is the set of palindromes in the alphabet $X^{\pm 1}$, $\mathcal{P}(Y)$ is the set of palindromes in the alphabet $Y^{\pm 1}$. If $\varphi : G \longrightarrow H$ be an epimorphism such that $\varphi(X) = Y$, then*

$$\text{pw}(H) \leq \text{pw}(G).$$

In [7] we also prove the following lemma.

Lemma 2.4. *Let $G = \langle X \rangle$ be a group generated by a set X . Then the following hold.*

- (1) *If p is a palindrome, then for any integer m the element p^m is also a palindrome.*
- (2) *Any element in G which is conjugate to a product of n palindromes, $n \geq 1$, is a product of n palindromes if n is even, and of $n + 1$ palindromes if n is odd.*

- (3) Any commutator of the type $[u, p]$, where p is a palindrome is a product of 3 palindromes. Any element $[u, x^\alpha]x^\beta$, $x \in X$, $\alpha, \beta \in \mathbb{Z}$, is a product of 3 palindromes.
- (4) Any commutator of the type $[u, pq]$, where p, q are palindromes is a product of 4 palindromes. Any element $[u, px^\alpha]x^\beta$, $x \in X$, $\alpha, \beta \in \mathbb{Z}$, is a product of 4 palindromes.

The following is a generalization of (1) and (4) in the above lemma.

Lemma 2.5. *Let $G = \langle X \rangle$ and $p_1, \dots, p_k$ are palindromes in G and $u \in G$. Then*

- (1) *If $q = p_1 p_2$ is a product of two palindromes, then for any integer m the element q^m is also a product of two palindromes.*
- (2)

$$l_{\mathcal{P}}([u, p_1 p_2 \dots p_k]) \leq 2k + \varepsilon \text{ where } \varepsilon = \begin{cases} 0 & \text{if } k \text{ is even,} \\ 1 & \text{if } k \text{ is odd.} \end{cases}$$

Proof. (1) Let $m \geq 1$. Then $q^m = q^{m-1} p_1 \cdot p_2$ is a product of two palindromes. If $m \leq -1$, then the assertion follows from the fact that the inverse to a palindrome is a palindrome.

(2) We see that

$$w = [u, p_1 p_2 \dots p_k] = u^{-1} (p_k^{-1} \dots p_2^{-1} p_1^{-1}) u p_1 p_2 \dots p_k.$$

By Lemma 2.4(2) the element $u^{-1} (p_k^{-1} \dots p_2^{-1} p_1^{-1}) u$ is a product of k palindromes if k is even and is a product of $k + 1$ palindromes if k is odd. Hence w is a product of $2k + \varepsilon$ palindromes. $\square$

Let $G = \langle X \rangle$ be a group. Let $\bar{G} = \langle \bar{X} \rangle$, where $\bar{X}$ is the image of X under some epimorphism of G . It follows from Lemma 2.3 that $\text{pw}(\bar{G}, \bar{X}) \leq \text{pw}(G, X)$. We now aim to find some upper bound for $\text{pw}(G, X)$.

Let H be a normal subgroup of G such that $G/H = \bar{G}$. Then H is generated by some elements $h_1, h_2, \dots$. The number of these elements may be infinite. However, any element $h \in H$ can be represented by some word in the alphabet $X^{\pm 1}$. Denote by $\text{pw}(H, X)$ the palindromic width of the subgroup H in the alphabet $X^{\pm 1}$.

Lemma 2.6. *Let $G = \langle X \rangle$ be a group generated by X . Let H be a normal subgroup of G and $\bar{G} = G/H = \langle \bar{X} \rangle$. Then*

$$\text{pw}(\bar{G}, \bar{X}) \leq \text{pw}(G, X) \leq \text{pw}(\bar{G}, \bar{X}) + \text{pw}(H, X).$$

Proof. It follows from Lemma 2.3 that $\text{pw}(\bar{G}, \bar{X}) \leq \text{pw}(G, X)$. To prove that $\text{pw}(G, X) \leq \text{pw}(\bar{G}, \bar{X}) + \text{pw}(H, X)$, suppose that $\text{pw}(\bar{G}, \bar{X}) = k$ and $\text{pw}(H, X) = l$. Any $g \in G$ is a product $g = h\bar{g}$, where $h \in H$ and $\bar{g}$ is the coset representative of G by H , i.e. $g \in H\bar{g}$. Since h is a product of at most k palindromes and $\bar{g}$ is a product of at most l palindromes, hence g is a product of at most $k + l$ palindromes. This proves the lemma. $\square$

Corollary 2.7. *Let $H = \langle Y \rangle$ be a group such that $\text{pw}(H, Y) < \infty$. Let $G = \langle X \cup Y \rangle$ be a finite extension of H . Then $\text{pw}(G, X \cup Y) < \infty$.*

Note that Lemma 2.6 is a palindromic version of [23, Lemma 1] where Rhemtulla proved similar inequalities for commutator widths. However, we do not know whether analogue of the above corollary holds for commutator widths. If $G = \langle X \rangle$ is a finitely generated group and G belongs to some variety $\mathcal{M}$ of groups, then to prove that $\text{pw}(G, X)$ finite, it follows from Lemma 2.3 that it is enough to prove $\text{pw}(F_n(\mathcal{M}))$ is finite.

In [7, Problem 2] we asked a question on the connection of the palindromic width and the commutator width of a group. The following proposition gives us this connection for some groups.

Proposition 2.8. *Let $G = \langle X \rangle$ be a finitely generated group with a set of generators $X = \{x_1, \dots, x_n\}$ of cardinality n . Suppose G has a non-trivial center $Z = Z(G)$. Let $\bar{G} = G/Z = \langle \bar{X} \rangle$. Let $\text{cw}(G) = l$ and $\text{pw}(\bar{G}, \bar{X}) = k$. Then*

$$\text{pw}(G, X) \leq n + l(2k + \varepsilon), \text{ where}$$

$$\varepsilon = \begin{cases} 0 & \text{if } k \text{ is even,} \\ 1 & \text{if } k \text{ is odd.} \end{cases}$$

Proof. We have the following exact sequence

$$1 \rightarrow Z \rightarrow G \xrightarrow{\pi} \bar{G} \rightarrow 1.$$

Let $s : \bar{G} \rightarrow G$ be a section. Then any element $g \in G$ has the form $g = z\tilde{g}$, where $z \in Z$ and $\tilde{g} = s(\bar{g})$ for some $\bar{g} \in \bar{G}$. Since $\text{cw}(G) = l$, any element $u \in G' = [G, G]$ has the form

$$u = [h_1, g_1][h_2, g_2] \dots [h_l, g_l], \quad h_i, g_i \in G \text{ for } i = 1, 2, \dots, l.$$

For $i = 1, 2, \dots, l$, write any element g_i in the form $g_i = z_i \tilde{g}_i$, $z_i \in Z$, $\tilde{g}_i = s(\bar{g}_i)$. Then

$$\begin{aligned} u &= [h_1, z_1 \tilde{g}_1][h_2, z_2 \tilde{g}_2] \dots [h_l, z_l \tilde{g}_l] \\ &= [h_1, \tilde{g}_1][h_2, \tilde{g}_2] \dots [h_l, \tilde{g}_l]. \end{aligned}$$

Note that arbitrary element $g \in G$ has the form

$$g = x_1^{\alpha_1} \dots x_n^{\alpha_n} [h_1, \tilde{g}_1][h_2, \tilde{g}_2] \dots [h_l, \tilde{g}_l],$$

for some integers $\alpha_1, \dots, \alpha_n$. By hypothesis, any element $\tilde{g}_i$ is a product of at most k palindromes and by Lemma 2.5(2), $[h_i, \tilde{g}_i]$ is a product of at most $2k + \varepsilon$ palindromes. Hence u is a product of at most $l(2k + \varepsilon)$ palindromes. Hence g is a product of at most $n + l(2k + \varepsilon)$ palindromes. This proves the proposition. $\square$

3. PALINDROMIC WIDTH OF SOLVABLE GROUPS: PROOF OF THEOREM 1.1

Lemma 3.1. *Let $G = \langle x_1, \dots, x_n \rangle$ be a finitely generated group and $H = \langle \langle y_1, \dots, y_k \rangle \rangle$ be the normal closure of elements $y_1, \dots, y_k$ in G . Then $\text{pw}(G/H') < \infty$ if and only if $\text{pw}(G/H'') < \infty$.*

Proof. Since G/H' is a homomorphic image of G/H'' , by Lemma 2.3, $\text{pw}(G/H') < \infty$ if $\text{pw}(G/H'') < \infty$.

Suppose that $\text{pw}(G/H') < \infty$. We must prove that $\text{pw}(G/H'') < \infty$. We may assume $H'' = 1$. Then by Lemma 2.6,

$$\text{pw}(G) \leq \text{pw}(G/[H', G]) + \text{pw}([H', G]).$$

By Lemma 2.1

$$[H', G] = \{[h_1, x_1][h_2, x_2] \dots [h_n, x_n] \mid h_i \in H'\}.$$

By Lemma 2.4(3) $\text{pw}([H', G]) \leq 3n$. Hence

$$\text{pw}(G) \leq \text{pw}(G/[H', G]) + 3n.$$

We need to show only that $\text{pw}(G/[H', G]) < \infty$.

We may now assume that $[H', G] = 1$. Thus $H' \leq Z(G)$ and H is a 2-step nilpotent group. Hence the map $h \mapsto [h, y_i]$, $h \in H$ is a homomorphism. Let

$$H_i = \{[h, y_i] \mid h \in H\}.$$

Then $H_i \leq H' \leq Z(G)$ and H_i is normal in G . So, $K = H_1 H_2 \dots H_k$ is a normal subgroup of G . Thus the centralizer in H of H/K :

$$Z_H(H/K) = \{h \in H \mid [h, H] \subset K\}$$

is also normal in G . However, $y_i \in Z_H(H/K)$ for all $i = 1, 2, \dots, k$. Hence H/K is abelian and $H' \leq K$. Also, we have $K \leq H'$. This implies, $K = H'$.

Every element of K can be written in the form

$$[h_1, y_1][h_2, y_2] \dots [h_k, y_k], \quad h_i \in H.$$

Any y_i is a word in the alphabet $X^{\pm 1}$. Let m be the maximal length $l_X(y_i)$ of y_i in the alphabet $X^{\pm 1}$ for all $i = 1, 2, \dots, k$ then by Lemma 2.5(2) we have $l_{\mathcal{P}}([h_i, y_i]) \leq 2m+1$. Hence $\text{pw}(H', X) \leq k(2m+1)$ and

$$\text{pw}(G) \leq \text{pw}(G/H') + k(2m+1).$$

This proves the lemma. $\square$

3.1. Proof of Theorem 1.1.

Proof. Let $X = \{x_1, x_2, \dots, x_n\}$. Take a minimal subset $Z \subseteq X$ such that $G = \langle A, Z \rangle$. Without loss of generality, assume that $Z = \{x_1, x_2, \dots, x_m\}$, $m \leq n$. Let $M = \langle Z \rangle$ so that $G = AM$. By Lemma 2.6,

$$(3.1) \quad \text{pw}(G, X) \leq \text{pw}(G/[A, G], \bar{X}) + \text{pw}([A, G], X),$$

where $\bar{X}$ is the image of X in $G/[A, G]$. It follows from Lemma 2.1 and Lemma 2.6 that

$$\text{pw}([A, G], X) \leq 3n.$$

Further, since we are considering $G/[A, G]$ in (3.1), without loss of generality we may further assume that $A \leq Z(G)$. Thus we have

$$G' = [AM, AM] = [M, M] = M'.$$

It is therefore sufficient to show that $\text{pw}(M, Z)$ is finite. Note that $M/M \cap A$ is isomorphic to G/A and it satisfies the maximal condition for normal subgroups. Also $A \cap M \leq Z(M)$.

We will now complete the proof by using induction on the solvability length of M . If $M \in \mathcal{U}$, i.e. M is abelian, then it follows from [7, Theorem 1] that $\text{pw}(M, Z) \leq m$. Suppose the result holds true for $M \in \mathcal{U}^r$. Consider the case when $M \in \mathcal{U}^{r+1}$. Let $L = M \cap A$. Since M/L satisfies the maximal condition for normal subgroups, there exists a subgroup $N = \langle\langle y_1, y_2, \dots, y_k \rangle\rangle$ of M such that N is the normal closure of elements $y_1, \dots, y_k$ of M and $LN = LM^{(r-1)}$. By induction $M^{(r+1)} = 1$, so $(LN)' = N' =$

$M^{(r)}$ is abelian. Now by Lemma 3.1, $\text{pw}(M/N') < \infty$ implies that $\text{pw}(M/N'') < \infty$. However, $N^{(2)} = M^{(r+1)} = 1$ and $M/N' \in \mathcal{U}^r$ and by hypothesis M/N' has a finite palindromic width. Hence the theorem follows. $\square$

4. PROOF OF THEOREM 1.2

Let $\mathcal{M}$ be some class of groups and $\mathcal{N}_c$ be the variety of nilpotent groups of nilpotency class $\leq c$. Clearly, for $c = 1$, $\mathcal{N}_1 = \mathcal{U}$. Let $\mathcal{L}_n$ be the class of groups generated by n elements. Denote by $\mathcal{L}_n \cap \mathcal{N}_c \mathcal{M}$ the class of groups that belong to both $\mathcal{L}_n$ and $\mathcal{N}_c \mathcal{M}$.

Lemma 4.1. *For any class $\mathcal{M}$ of groups and any n , every group in $\mathcal{L}_n \cap \mathcal{N}_2 \mathcal{M}$ has finite palindromic width if and only if every group in $\mathcal{L}_n \cap \mathcal{N}_c \mathcal{M}$ has finite palindromic width for $c \geq 2$.*

Proof. If every group in $\mathcal{L}_n \cap \mathcal{N}_c \mathcal{M}$, $c \geq 2$, has finite palindromic width, then evidently every group in $\mathcal{L}_n \cap \mathcal{N}_2 \mathcal{M}$ has finite palindromic width. We will use induction on c to prove the converse. If $c = 2$ then there is nothing to prove. Assume the result holds for $c = m - 1$. Let $G \in \mathcal{L}_n \cap \mathcal{N}_m \mathcal{M}$. Hence there is a normal subgroup N of G such that $G/N \in \mathcal{M}$ and $N \in \mathcal{N}_m$. Thus $\gamma_{m+1}(N) = 1$. Since $m > 2$, $\gamma_m(N) = [\gamma_{m-1}(N), G]$ is abelian and hence by Lemma 2.6

$$\text{pw}(G) \leq \text{pw}(G/[\gamma_{m-1}(N), G]) + n.$$

Since $G/[\gamma_{m-1}(N), G] \in \mathcal{L}_n \cap \mathcal{N}_{m-1} \mathcal{M}$, hence $\text{pw}(G) < \infty$. This proves the lemma. $\square$

4.1. Proof of Theorem 1.2.

Proof. Let G be a finitely generated group in $\mathcal{U} \mathcal{N} \mathcal{N}$. Then for some integers m, l , $\gamma_m(\gamma_l(G)) = A$ is abelian and normal in G . Thus by Lemma 2.6 and Lemma 2.1,

$$\text{pw}(G) \leq \text{pw}(G/[A, G]) + 3n,$$

where n is the number of generators of G . Now

$$\gamma_{m+1}(\gamma_l(G)) = [A, \gamma_l(G)] \subset [A, G],$$

so that $G/[A, G] \in \mathcal{L}_n \cap \mathcal{N}_m \mathcal{N}_{l-1}$. It is sufficient to show that any finitely generated group in $\mathcal{N} \mathcal{N}$ has finite palindromic width. By Lemma 4.1, it is enough to show that any group in the class $\mathcal{L}_n \cap \mathcal{N}_2 \mathcal{N}$ has finite palindromic width.

We may now suppose that $G \in \mathcal{L}_n \cap \mathcal{N}_2 \mathcal{N}$ so that G contains a normal subgroup $M \in \mathcal{N}_2$ and $G/M \in \mathcal{N}$. Now G/M' is finitely generated group in the class $\mathcal{U} \mathcal{N}$. We know from [7, Proposition 3.7] that the palindromic width of these groups is bounded above by $5n$. Furthermore M' is abelian and normal in G and consequently has finite palindromic width. Hence G has finite palindromic width. $\square$

5. PALINDROMIC AND COMMUTATOR WIDTHS OF WREATH PRODUCTS $\mathbb{Z} \wr \mathbb{Z}$

We have proven that if $F_n(\mathcal{U}^2)$ is a free metabelian group of rank n , then $\text{pw}(F_n(\mathcal{U}^2)) \leq 4n - 1$. We think that this boundary is rough. For some metabelian groups we can find the faithful value of palindromic width. For example, if

$$BS(1, n) = \langle a, t \mid t^{-1}at = a^n \rangle, \quad n \in \mathbb{Z} \setminus \{0\}$$

is a metabelian Baumslag-Solitar group, then we can prove

Proposition 5.1. $\text{pw}(BS(1, n), \{a, t\}) = 2$.

Proof. It is not difficult to prove (see for example [6, Lemma 11]) that every element $g \in BS(1, n)$ can be presented in the form $g = t^k a^l t^{-m}$ for some non-negative integers k, m and some integer l . Then

$$g = t^k a^l t^k \cdot t^{-m-k}$$

is the product of 2 palindromes. On the other side it is not difficult to prove, that element ta is not a palindrome. $\square$

In this section, we aim to obtain palindromic width of $G = \mathbb{Z} \wr \mathbb{Z}$ where the palindromes are considered with respect to the generating set $\{a^{\pm 1}, b^{\pm 1}\}$, where a is the generator of the first cyclic group in the product and b is the generator of the second cyclic group. Let $A = \langle a \rangle \cong \mathbb{Z}$, $B = \langle b \rangle \cong \mathbb{Z}$. Then

$$G = \mathbb{Z} \wr \mathbb{Z} = A \wr B.$$

Let $a_0 = a$ and $A_0 = A = \langle a_0 \rangle$. Define the infinite set of groups $A_i = \langle a_i \rangle$ which are isomorphic copies of A and $i \in \mathbb{Z}$. Let $C = \prod_{i \in \mathbb{Z}} A_i$ be the direct products of the A_i 's. We can identify the group C as a subgroup of the group $F(B, A)$ of functions from B to A , where any function sends the element b^i onto some element in A_i . If $f \in C \subset F(B, A)$, then we will write

$$f = \prod_{i \in \mathbb{Z}} a_i^{n_i}, \quad n_i \in \mathbb{Z}, \quad f(b^i) = f(i) = a_i^{n_i},$$

where only finitely many numbers n_i are non-zero. The group B acts on C by the rule

$$f^b = \prod_{i \in \mathbb{Z}} a_{i+1}^{n_i}.$$

This action induces an automorphism of C . Then $G = A \wr B = C \rtimes B$, where $\rtimes$ denotes the semi-direct product.

For any $f \in C$, define $\text{supp}(f)$ to be the set of indices i such that $n_i \neq 0$. For any $f \in C$, $\text{supp}(f)$ is a finite set. Also we have the following conjugation rules for arbitrary integers k and l :

$$b^{-k} A_l b^k = A_{l+k}.$$

5.1. Commutator Subgroup of G .

Lemma 5.2. *The commutator subgroup $G' = [G, G]$ is equal to the group $[C, B]$, where $[C, B]$ is a subgroup of G generated by the commutators $[c, b^l]$ for $c \in C$, $l \in \mathbb{Z}$.*

Proof. The inclusion $[C, B] \subset [G, G]$ is evident. To prove the reverse inclusion, consider any arbitrary commutator $[cb^k, c'b^l]$ in $[G, G]$, where $c, c' \in C$, $k, l \in \mathbb{Z}$. Then

$$\begin{aligned} [cb^k, c'b^l] &= b^{-k} c^{-1} b^{-l} (c')^{-1} c b^k c' b^l \\ &= (c^{-1})^{b^k} b^{-k} b^{-l} (c')^{-1} c b^k b^l (c')^{b^l} \\ &= c_k^{-1} (c')^{-b^{k+l}} c^{b^{k+l}} (c')^{b^l}, \quad \text{where } c_k = b^{-k} c b^k \\ &= c_k^{-1} (c'_{k+l})^{-1} c_{k+l} c'_l \\ &= c_k^{-1} c_{k+l} (c'_{k+l})^{-1} c'_l = [c_k, b] [c'_{k+l}, b^{-k}]. \end{aligned}$$

Hence $[G, G] \subset [C, B]$. This proves the lemma. $\square$

We represent $c \in G$ as a product

$$(5.1) \quad c = \prod_{i \in \mathbb{Z}} a_i^{n_i}, \quad n_i \in \mathbb{Z},$$

or as function $c : \mathbb{Z} \rightarrow \mathbb{Z}$ given by $c(i) = n_i$. Define a function $\log : C \rightarrow \mathbb{Z}$ as

$$\log c = \sum_{i \in \mathbb{Z}} n_i.$$

From Lemma 5.2 we have the following.

Corollary 5.3. *The commutator subgroup G' is contained in C . An element $c \in C$ belongs to G' if and only if $\log c = 0$.*

Lemma 5.4. *Any element in the commutator subgroup G' is a commutator.*

Proof. Let $c \in G'$. We want to find an element $f \in C$ be such that $[f, b] = c$. To do this denote by m the minimal integer m such that $n_m \neq 0$ in the representation of (5.1) and by M the maximal integer M such that $n_M \neq 0$ in (5.1). Hence the element c has the form

$$c = a_m^{n_m} a_{m+1}^{n_{m+1}} \cdots a_{m+s-1}^{n_{m+s-1}} a_{m+s}^{n_{m+s}},$$

where $M = m + s$ and by Corollary 5.3,

$$(5.2) \quad n_m + n_{m+1} + \cdots + n_{m+s-1} + n_{m+s} = 0.$$

We will find the element f in the form

$$f = a_m^{x_m} a_{m+1}^{x_{m+1}} \cdots a_{m+s-1}^{x_{m+s-1}}$$

with unknowns $x_m, x_{m+1}, \dots, x_{m+s-1}$. Then the commutator

$$\begin{aligned} [f, b] = f^{-1} f^b &= a_m^{-x_m} a_{m+1}^{-x_{m+1}} \cdots a_{m+s-1}^{-x_{m+s-1}} \cdot a_m^{x_m} a_{m+1}^{x_{m+1}} \cdots a_{m+s-1}^{x_{m+s-1}} a_{m+s}^{x_{m+s}} \\ &= a_m^{-x_m} a_{m+1}^{x_m - x_{m+1}} a_{m+2}^{x_{m+1} - x_{m+2}} \cdots a_{m+s-1}^{x_{m+s-2} - x_{m+s-1}} a_{m+s}^{x_{m+s-1}}. \end{aligned}$$

Note that the equation $[f, b] = c$ gives us the following system of equations:

$$\begin{aligned} -x_m &= n_m \\ -x_{m+1} + x_m &= n_{m+1} \\ -x_{m+2} + x_{m+1} &= n_{m+2} \\ &\vdots \\ -x_{m+s-1} + x_{m+s-2} &= n_{m+s-1} \\ x_{m+s-1} &= n_{m+s}. \end{aligned}$$

From the first s equations we find

$$\begin{aligned} x_m &= -n_m \\ x_{m+1} &= x_m - n_{m+1} = -n_m - n_{m+1}, \\ x_{m+2} &= x_{m+1} - n_{m+2} = -n_m - n_{m+1} - n_{m+2}, \\ &\vdots \\ x_{m+s-1} &= x_{m+s-2} - n_{m+s-1} = -n_m - n_{m+1} - \cdots - n_{m+s-1}. \end{aligned}$$

Inserting this value of x_{m+s-1} in the last equation we have

$$-n_m - n_{m+1} - \cdots - n_{m+s-1} = n_{m+s},$$

or

$$n_m + n_{m+1} + \cdots + n_{m+s-1} + n_{m+s} = 0.$$

Since $c \in G'$ we see in (5.2) that this equality holds. So, we have a well-defined solution to the above system of equations and f exists in C such that $[f, b] = c$. This proves the theorem. $\square$

Corollary 5.5. $\text{pw}(G, \{a, b\}) \leq 3$.

Proof. Every element in G can be represented in the form

$$g = a^k b^l d, \quad d \in G', k, l \in \mathbb{Z}.$$

By Theorem 5.4, there exists an element $f \in C$ such that $d = [f, b]$. Hence

$$g = a^k b^l [f, b] = a^k [f^{b^{-l}}, b] b^l.$$

Since elements in C are commute then

$$\begin{aligned} g &= a^k [f^{b^{-l}}, b] b^l = a^k f^{-b^{-l}} b^{-1} f^{b^{-l}} b^{l+1} = a^k f^{-b^{-l}} b^{-1} \left(\overline{f^{-b^{-l}}} \overline{f^{b^{-l}}} \right) f^{b^{-l}} b^{l+1} = \\ &= f^{-b^{-l}} b^{-1} \overline{f^{-b^{-l}}} \cdot \overline{f^{b^{-l}}} a^k f^{b^{-l}} \cdot b^{l+1}. \end{aligned}$$

Hence g is a product of at most 3 palindromes. $\square$

5.2. Palindromes in G . The following proposition describes palindromes in G .

Proposition 5.6. *The palindromes in G have one of the following forms:*

$$a_{-k}^l f f^{b^{-2k}} b^{2k} \text{ or } f f^{b^{-l}} b^l, \quad k, l \in \mathbb{Z}.$$

Proof. Since any element in G has the form $f b^k$, $f \in C$, $k \in \mathbb{Z}$, then any palindrome has one of the following forms

$$f b^k a^l \overline{f b^k}, \quad f b^k b^l \overline{f b^k}.$$

Since C is an abelian group, it follows that $\overline{f b^k} = b^k f$. Hence

$$f b^k a^l \overline{f b^k} = f b^k a^l b^k f = f (a^l)^{b^{-k}} b^{2k} f = f a_{-k}^l f^{b^{-2k}} b^{2k} = a_{-k}^l f f^{b^{-2k}} b^{2k},$$

and

$$f b^k b^l \overline{f b^k} = f b^{2k+l} f = f f^{b^{-(2k+l)}} b^{2k+l}.$$

This proves the lemma. $\square$

We will call palindromes of the first type by *a-palindromes* and the palindromes of the second kind by *b-palindromes*. We see that we can find elements in G that are not a product of *a-palindromes*, nor *b-palindromes*. It would be interesting to verify whether every element of G is a product of an *a-palindrome* and a product of a *b-palindrome*.

Theorem 5.7. $\text{pw}(G, \{a, b\}) = 3$.

The theorem will follow from the following proposition, Lemma 2.3 and Corollary 5.5.

Proposition 5.8. *There is a homomorphism of the group $\mathbb{Z} \wr \mathbb{Z}$ onto $N_{2,2}$.*

Proof. The group $N_{2,2}$ has the following presentation:

$$(5.1) \quad N_{2,2} = \langle x, y \mid [y, x, x] = [y, x, y] = 1 \rangle.$$

Consider the quotient $G/\gamma_3(G)$. We use the same letters a and b for the images of a and b in $G/\gamma_3(G)$. Hence we have the following relations in $G/\gamma_3(G)$:

$$(5.2) \quad [b, a, a] = 1, [b, a, b] = 1.$$

Since $[b, a] = b^{-1}a^{-1}ba = a_1^{-1}a = a_1^{-1}a_0$, we have $[b, a, a] = [a_1^{-1}a_0, a_0] = 1$, which is the first relation in (5.2). The second relation has the form

$$[b, a, b] = [a_1^{-1}a_0, b] = a_0^{-1}a_1ba_1^{-1}a_0b = a_0^{-1}a_1a_2^{-1}a_1 = a_0^{-1}a_1^2a_2^{-1} = 1,$$

i.e.

$$(5.3) \quad a_0^{-1}a_1^2a_2^{-1} = 1.$$

We can write a_2 in the form

$$a_2 = a_0^{-1}a_1^2.$$

Conjugating this relation by b we have

$$a_3 = a_1^{-1}a_2^2 = a_1^{-1}(a_0^{-1}a_1^2)^2 = a_0^{-2}a_1^3.$$

Conjugating this relation by b we have

$$a_4 = a_1^{-2}a_3^3 = a_1^{-2}(a_0^{-1}a_1^2)^3 = a_0^{-3}a_1^4.$$

Continuing this process we can remove all generators $a_2, a_3, \dots, a_k, \dots$ in the image of the group C in $G/\gamma_3(G)$. On the other hand, conjugating the relation (5.3) by b^{-1} gives

$$a_{-2} = a_2^2a_0^{-1} = (a_0^2a_1^{-1})^2a_0^{-1} = a_0^3a_1^{-2}.$$

Continuing this process we can remove all generators $a_{-1}, a_{-2}, \dots, a_{-k}, \dots$ in the image of C in $G/\gamma_3(G)$. Hence, any element g in $G/\gamma_3(G)$ has the form

$$g = a_0^k a_1^l b^m = a^k b^{-1} a^l b b^m = a^k a^l (a^{-l} b^{-1} a^l b) b^m = a^{k+l} [a^l, b] b^m.$$

Since in $G/\gamma_3(G)$, commutator $[a^l, b] = [a, b]^l$ commutes with b^l , thus

$$g = a^{k+l} b^m [b, a]^{-l}.$$

This shows that $G/\gamma_3(G)$ is isomorphic to $N_{2,2}$. This proves the proposition. $\square$

6. OPEN QUESTIONS

We proved that for group G in Theorems 1.1 and 1.2 the palindromic width $\text{pw}(G)$ is finite.

Question 1. Let G be a group in Theorems 1.1 and 1.2. Find the lower and upper bounds on the palindromic width $\text{pw}(G)$.

In the last section we proved that the palindromic width of a solvable Baumslag-Solitar group is equal 2. For the non solvable groups we can formulate the following.

Question 2. Let

$$BS(m, n) = \langle a, t \mid t^{-1}a^mt = a^n \rangle, \quad m, n \in \mathbb{Z} \setminus \{0\}$$

be a non solvable Baumslag-Solitar group. Is it true that $\text{pw}(BS(m, n), \{a, t\})$ is infinite?

Also, we can formulate the general question on the palindromic width of HNN-extensions.

Question 3. *Let $G = \langle X \rangle$ be a group and A and B are proper isomorphic subgroups of G and $\varphi : A \rightarrow B$ be an isomorphism. Is it true that the HNN-extension*

$$G^* = \langle G, t \mid t^{-1}At = B, \varphi \rangle$$

of G with associated subgroups A and B has infinite palindromic width with respect to the generating set $X \cup \{t\}$?

In [10] palindromic widths of free products were investigated. For the generalized free products we can formulate:

Question 4. *Let $G = A *_C B$ be a free product of A and B with amalgamated subgroup C and $|A : C| \geq 3$, $|B : C| \geq 2$. Is it true that $\text{pw}(G, \{A, B\})$ is infinite?*

It would be interesting to understand palindromic width of wreath products. In particular, one can ask the following.

Question 5. *Let $G = A \wr B$ be a wreath product of group $A = \langle X \rangle$ and $B = \langle Y \rangle$ such that $\text{pw}(A, X) < \infty$ and $\text{pw}(B, Y) < \infty$. Is it true that $\text{pw}(G, X \cup Y) < \infty$?*

Riley and Sale [24] have proved that if B is a finitely generated abelian group then the answer is positive. Fink [14] has also proved the finiteness of palindromic width of wreath product in for some more cases. However, the general case when B is a finitely generated non-abelian group is still open.

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